Lex Cheat Sheet

1 Labels

Available levels:
law, title, chapter, section, article, paragraph, point, subpoint

Levels required for all rules:
law, article

Example:

```
    law "Name of law"
    chapter "A" "Optional title of chapter"
    article "6" "Optional title of article"
    paragraph "1"
    point "1"
    ...
    article "7"
```

2 Types

Base types:
bool, int, float, string, span, time, money c

Type declarations:

```
type user_consent_action

type user_id
    """a user identifier"""

type purpose is string
    """a purpose of processing"""
```

Possible values:

```
true, false                         # bool
1, 18, 42                           # int
1.2, 3.1415, 4.                     # float
"hello", "Erika Mustermann"         # string
2s, 3m, 4h, 6d, 7M, 8y              # span
‘2024-01-01 08:32:27‘               # time
|CHF 2.|, |CHF 4000.25|             # money CHF
```

3 Events

Two keywords:
event (for actions), predicate (for relations)

Capabilities:
observable < suppressable, causable, causable observable, causable suppressable, internal

Non-internal → ‘brute facts’ in the world vs. internal → ‘institutional facts’ in the law.

Event declarations:

```
observable predicate PersonalData
    """data {d} is personal data of data
     subject {ds}"""
    d : data
    ds : data_subject

suppressable event ProcessesData
    """company {c} processes data {d} for
     purpose {p} as part of data processing
     activity {a}"""
    c : company
    d : data
    p : purpose
    a : activity

internal event IsLegal
    """processing activity {a} is legal"""
    a : activity
```

4 Expressions

`( ... )`	Brackets
`true`, `false`, 1, 2.5, ...	Constants
`=`, `<>`, `<`, `>`, `<=`, `>=`	Relations
`+`, `-`, `*`, `/`, `^`	Arithmetics
`Event(`t_1`,...,`t_k`)`	Event/Predicate
`NOT`	Negation
`AND`, `OR`, `IMPLIES`, `IFF`	Connectives
`EXISTS x.`	There exists x...
`FORALL x.`	For all x...
`ONCE`	‘some time in the past...’
`HISTORICALLY`	‘at all times in the past...’
`EVENTUALLY`	‘some time in the future...’
`ALWAYS`	‘always in the future...’

Temporal connectives can be followed by intervals `[a, b]`, `[a, b)`, `(a, b]`, or `(a, b)` where `b` can be `*` (∞). Bounds are integers followed by an optional time unit in `s`, `m`, `h`, `d`, `M`, `Y` (default: `s`).

5 Rules

Obligation rules: *‘whenever all of the premisses are true, then all the conclusions shall be true’*
Syntax: whenever... oblige

```
# [GDPR Art. 5] Personal data shall be:
# 1. processed lawfully, fairly and in a
# transparent manner in relation to the
# data subject

article "5"
paragraph "1"
rule
    whenever
        ProcessesData(c, d, p, a)
    oblige
        IsLawful(a)
        IsFair(a)
        IsTransparent(a)
```

After this rule, we can add either of the two lines

```
enforceable suppressing ProcessesData
enforceable causing IsLawful, IsFair,
     IsTransparent
```

Constitutive rules: *‘whenever all the premisses are true, then all the conclusions are the case (by definition)’*
The conclusions **must be** internal events!
Syntax: whenever... constitute

```
# Processing for a purpose is lawful when the
# data subject has given consent for
# processing their data for that purpose
# (consent was given some time in the past)

rule
    whenever
        ProcessesData(c, d, p, a)
        PersonalData(d, ds)
        ONCE GivesConsent(ds, c, p)
    constitute
        IsLawful(a)
```

Exception rules: *‘whenever all the premises are true, then the rules in the conclusions shall not apply’*
The conclusions **must be** labels!
Syntax: whenever... except

```
# Art. 5 par. 1 shall not apply when there
# exists a legal warrant to use the data

rule
    whenever
        LegalWarrant(c, d)
    except
        article "5" paragraph "1"
```